FIITJEE
ALL INDIA TEST SERIES
JEE (Advanced)-2024
FULL TEST – XI
PAPER –2
TEST DATE: 19-05-2024

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 51 questions.
- Each subject (PCM) has 17 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Three Sections**: **Section-A, Section-B & Section-C**.
  Section – A (01 – 04, 18 – 21, 35 – 38): *This section contains **TWELVE (12)** questions. Each question has **FOUR** options. **ONLY ONE** of these four options is the correct answer.*
  Section – A (05 –07, 22 – 24, 39 – 41): *This section contains **NINE (09)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).*
  Section – B (08 – 13, 25 – 30, 42 – 47): *This section contains **EIGHTEEN (18)** numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*
  Section – C (14 –17, 31 – 34, 48 – 51): *This section contains **SIX (06) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

MARKING SCHEME

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	If three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

Section – C: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct integer is entered;
Zero Marks	:	0	Question is unanswered;
Negative Marks	:	0	In all other cases.

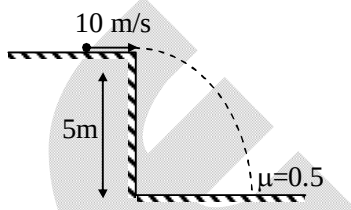
Physics

PART – I

SECTION – A
(One Options Correct Type)

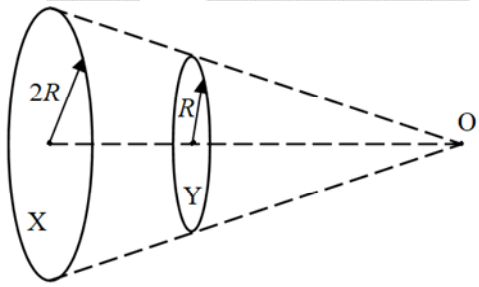
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

1. A small ball moving with a velocity 10 m/s, horizontally (as shown in figure) strikes a rough horizontal surface having $\mu = 0.5$. If the coefficient of restitution is *e* = 0.4. Horizontal component of velocity of ball after first impact will be (g = 10 m/s^2)



(A) 10 m/s (B) 8 m/s
(C) 3 m/s (D) 4 m/s

2. Two coaxial circular coils X and Y having equal number of turns and carrying equal currents in same sense. They subtend the same solid angle at point O. Let *B* be the magnetic field at O due to smaller coil Y, then



(A) Magnetic field at O due to larger coil is $\frac{B}{2}$
(B) Magnetic field at O due to larger coil is *B*
(C) Total magnetic field at O is 1.5*B*
(D) Data insufficient to find net magnetic field at O

3. In a non-metallic sphere of uniform surface charge density σ, a point P is given as shown in the figure(a). Nearest distance of P from the surface is infinitesimally small. Now a hemisphere is cut and struck to the other half as shown in the figure(b). If sticking has no effect on charge distribution, then electric field at P due to this arrangement is $\frac{n\sigma}{3\varepsilon_0}$. Find *n*.

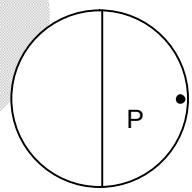


Fig (a)

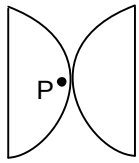
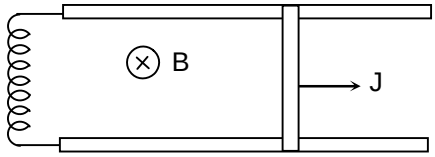


Fig (b)

(A) 3 (B) 2
(C) 1 (D) 0

4. Two parallel resistance less rails are connected by an inductor of inductance L at one end as shown in the figure. A magnetic field B exists in the space which is perpendicular to the plane of the rails. Now a conductor of length ℓ and mass *m* is placed transverse on the rails and given an impulse J towards the rightward direction. Then choose the correct option(s).



(A) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{4B^2\ell^2m}}$

(B) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{2Lm}}$

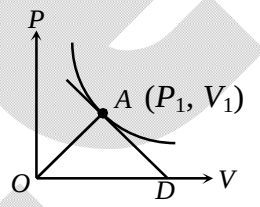
(C) Velocity of the conductor is half of the initial velocity after a displacement of the conductor $d = \sqrt{\frac{3J^2L}{B^2\ell^2m}}$

(D) Current flowing through the inductor at the instant when velocity of the conductor is half of the initial velocity is $i = \sqrt{\frac{3J^2}{mL}}$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

5. *n* moles of an ideal gas undergo an isothermal process at temperature *T*. P-V graph of the process is as shown in the figure. A point *A* (V_1, P_1) is located on the P-V curve. Tangent at point *A*, cuts the *V*-axis at point *D*. *AO* is the line joining the point *A* to the origin *O* of PV diagram. Then,



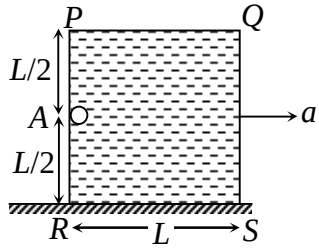
(A) coordinates of points *D* is $\left(\frac{3V_1}{2}, 0\right)$

(B) coordinates of points *D* is $(2V_1, 0)$

(C) area of the triangle *AOD* is *nRT*

(D) area of the triangle *AOD* is $\frac{3}{4}nRT$

6. A small solid ball of density ρ is held inside at point *A* a cubical container of side *L*, filled with an ideal liquid of density 4ρ as shown in the figure. Now, if the container starts moving with constant acceleration *a* horizontally and the ball is released from point *A* simultaneously, then



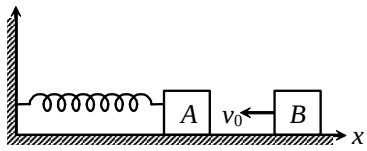
(A) For ball to hit the top of container at end *Q*, *a* = 3g

(B) For ball to hit the top of container at end *Q*, *a* = 2g

(C) Ball hits the top of container at end *Q* after a time *t* = $\sqrt{\frac{L}{3g}}$

(D) Ball hits the top of container at end *Q* after a time *t* = $\sqrt{\frac{2L}{3g}}$

7. A block *A* of mass *m* connected with a spring of force constant *k* is executing SHM. The position (*x*) and time (*t*) equation of the block is $x = x_0 + a\sin\omega t$. An identical block *B* moving towards negative *x*-axis with velocity v_0 collides elastically with block *A* at time *t* = 0. Then



(A) displacement time equation of *A* after collision will be $x = x_0 - v_0\sqrt{\frac{m}{k}}\sin\omega t$

(B) displacement time equation of *A* after collision will be $x = x_0 + v_0\sqrt{\frac{m}{k}}\sin\omega t$

(C) velocity of *B* just after collision will be *a* ω towards positive *x*-direction

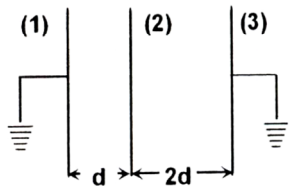
(D) velocity of *B* just after collision will be v_0 towards positive *x*-direction

SECTION – B

(Numerical Answer Type)

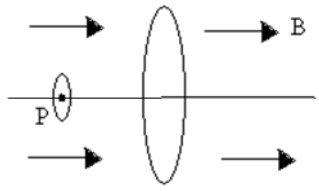
*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

8. Three identical plates of area *A* are arranged as shown. A charge +*Q* is given to plate (2). The energy stored between the plates (1) and (2) is $\frac{aQ^2d}{b\varepsilon_0 A}$, where *a* and *b* are relatively prime integers. Find $|a-b|$.

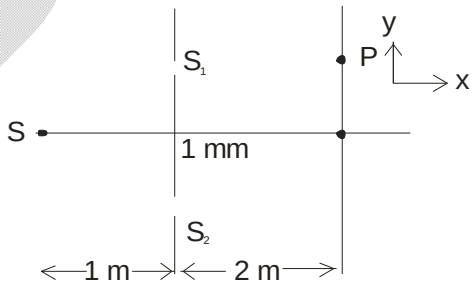


9. A wire of length L and 3 identical cells of negligible internal resistances are connected in series. Due to the current the temperature of the wire is raised by ΔT in a time t. A number N of similar cells are now connected in series with a wire of the same material and cross-section but of length 2L. The temperature of the wire is raised by the same amount ΔT in the same time t. find the value of N.

10. Figure shows a convex lens of focal length 12 cm lying in a uniform magnetic field B of magnitude 1.2 T parallel to its principle axis. A particle having a charge 2×10^{-3} C and mass 2×10^{-5} kg is projected perpendicular to the plane of the diagram with a speed of $4.8\,ms^{-1}$. The particle moves along a circle with its centre on the principal axis at a distance of 18 cm from the lens. The image of the particle is circle about the axis. The radius of the circle of image is _____cm.



11. In a Young's double slit experiment set up, source S of wavelength 500 nm illuminates two slits S_1 and S_2 which act as two coherent sources in phase with each other. The source S oscillates across the central line according to the equation $y = 0.5\sin\left(\frac{\pi t}{2}\right)$ where *y* is in mm and *t* in seconds. Given at t = 0 source is on central line. The minimum value of time *t* at which the intensity at point P on the screen exactly in front of the upper slit becomes minimum is



12. What should be the approximate K.E. of an electron so that its de-Broglie wavelength is equal to the wavelength of x-ray of maximum energy produced in an x-ray tube operating at 24,800 V? ($h = 6.6\times10^{-34}$ J-sec, mass of electron $m = 9.1\times10^{-31}$ kg) (give answer in 10^2 eV).

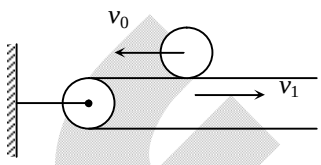
13. Two spherical mirrors each of same radii of curvature *R* one convex and other concave are placed facing each other. They are co-axial at a distance 2*R* from each other. A small portion in the form of circle of radius *a* is cut out from convex mirror near the pole. The radius of the second image of the hole is $\frac{a}{w2^3+x2^2+y2^1+z2^0}$; $w,x,y,z\in[0,1]$ where w, x, y , z are integers. Find the value of $w+2(x+y+z)$.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 14 and 15

A sphere of radius r and mass m has a linear velocity v_0, directed to the left and no angular velocity as it is placed on a belt at $t = 0$. The belt is moving to the right with a constant velocity $v_1 = 4$ m/s. After sliding on the belt for some distance the sphere starts pure rolling on the belt and is found to have no linear velocity of its centre of mass with respect to ground. Coefficient of friction between the belt and the sphere is $\mu_k = 0.2$.



14. The time t_1 (in sec), when the sphere will start pure rolling is …….

15. The distance (in meter) moved by sphere relative to belt in time t_1 is …….

Paragraph for Question Nos. 16 and 17

Incident wave $y = A\sin\left(ax + bt + \frac{\pi}{2}\right)$ is reflected by an obstacle at $x = 0$ which reduces intensity of reflected wave by 36%. Due to superposition, the resulting wave consists of a standing wave and a traveling wave formed which is given by

$$y = -1.6A\sin ax \cdot \sin bt + cA\cos(bt + ax)$$

where A, a, b, c are positive constants.

In this resultant wave maximum speed of particle is v.

16. Value of c is …….

17. Maximum velocity v of particle is kbA. Find the value of k.

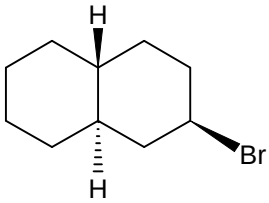
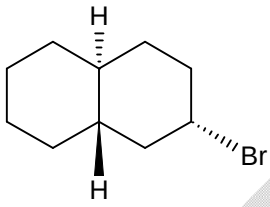
Chemistry

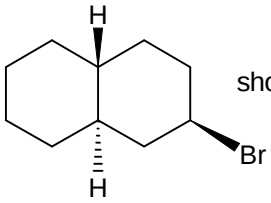
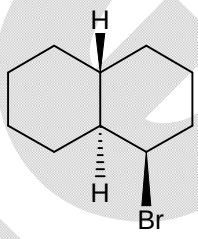
PART – II

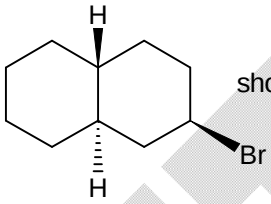
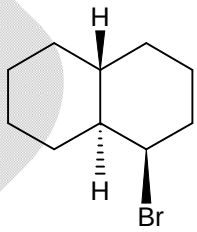
SECTION – A
(One Options Correct Type)

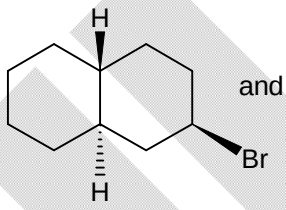
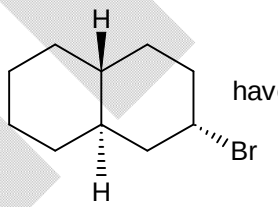
*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

18. Choose the correct statement(s) among the following

I.  and  have similar reactivity towards E2 reaction

II.  shows faster reactivity than  towards E2 reaction

III.  shows slower reactivity than  towards E2 reaction

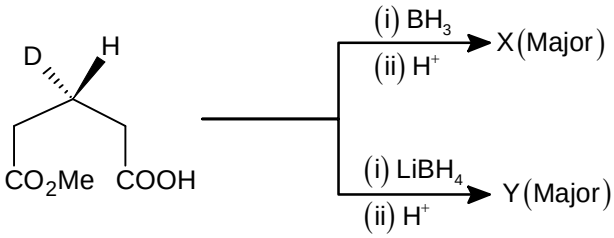
IV.  and  have simialr reactivity towards E2 reaction

(A) I, II, III and IV
(B) I and III only
(C) I, II and III only
(D) II and IV only

19. Choose the **incorrect** option
(A) Due to Schottky defect, density of crystal decreases.
(B) Number of next nearest neighbours of Cs^+ ion in CsCl structure is 6 Cs^+ ion.
(C) ABAB…… type of close packing gives rise to cubic close packing (ccp) lattice.
(D) Orthorhombic crystal may have end centred unit cell.

20. Liquid range of solvent is defined as the temperature range in which solvent exist in liquid state. According to above statement which of the following solvent has highest value of liquid range
(A) H_2O
(B) NH_3
(C) H_2SO_4
(D) HF

21.



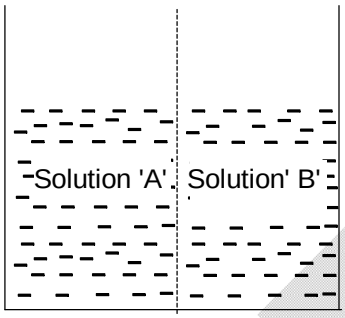
Correct relationship between X and Y is:

(A) Homomers
(B) Structural isomers
(C) Diastereomers
(D) Enantiomers

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

22. Solution 'A' containing 0.01 M $FeCl_3(aq)$ and solution 'B' containing 0.1 M $K_4\left[Fe(CN)_6\right](aq)$ are separated by semipermeable membrane as shown below at 300 K, prussian blue coloured solution is obtained when solution 'A' and solution 'B' are mixed together.



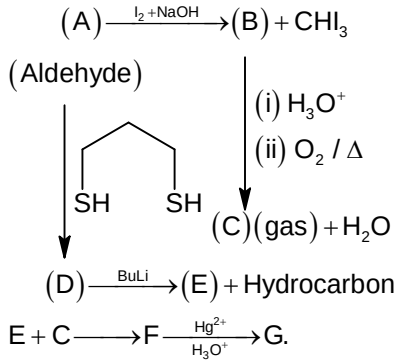
Which of the following option is/are correct considering $K_4\left[Fe(CN)_6\right]$ ionizes 60% and $FeCl_3$ ionises 80% in aq. solution.

(A) Solution A is hypertonic
(B) Blue colour will from in Solution A side
(C) Blue colour will from in Solution B side
(D) Initially Solution B will have 10 times higher osmotic pressure than that of Solution A.

23. One mole of an ideal gas expanded reversibly and isothermally at 27°C to double of its initial volume and then heated reversibly at constant volume from 27°C to 327°C. Total change in entropy will be

(A) $(R - C_V)\ell n2$
(B) $(R + C_V)\ell n2$
(C) $R\ell n2 + C_V \ell n\left(\frac{327}{27}\right)$
(D) $C_P \ell n2$

24. Consider the following sequence of reactions and choose the correct option(s)



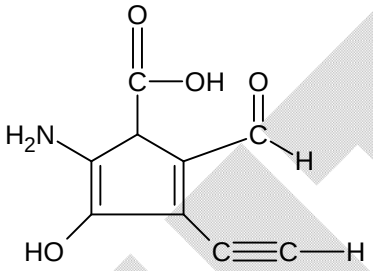
(A) The gas 'C' is a acidic oxide.
(B) The molecular weight of 'G' is 88 g/mol.
(C) CO_2 gas produced when compound 'G' is treated with $NaHCO_3$.
(D) Compound 'G' can be prepared by directly oxidation of 'A'.

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

25. 0.84 mole of compound 'A' is reacted with excess CH_3MgBr, after the completion of reaction the gas produced is used as a fuel to heat 50 kg H_2O at 25°C. What will be the change in temperature if all the fuel is consumed.



(A)

Considering all heat is used to increase the temprature of water only.

Given: specific heat of the water is 4.2 $Jgm^{-1}K^{-1}$. Heat of combustion of CH_4 is 900 kJ/mol

26. The osmotic pressure of 0.1 M aq. solution of weak monobasic acid at 300 K in found to be 3 atm. On increasing temperature from 300 K to 360 K the osmotic pressure is increased by 40%. What will be percentage change in degree of dissociation due to increase in temperature.
$\left(\text{Given}: R = \frac{1}{12}\text{lit atm } K^{-1} \text{ mol}^{-1}\right)$ [Assume molecularity of solution doesn't change by changing temperature]

27. Find total number of species having "Bent or V shape".
I_3^-, $BeCl_2$, NH_2^-, OH_2, XeF_2, N_2O, SO_2, SF_2

28. A metal X exists as X^{+2} in a non-chelating complex having effective atomic number (EAN) 34, find the number of monodentante ligands in complex, where atomic weight of metal is 28.

29. The conductivity of 20 mL of 0.2 N NaOH solution is 2.2×10^{-2} Scm^{-1}. If 10 mL of 0.4 N HCl solution is added to this solution, the conductivity decreases by 2×10^{-2} Scm^{-1}. The equivalent conductivity of the final solution in $Scm^2\,eq^{-1}$ is …….

30. Consider the following data for the reaction

$A + B \longrightarrow P$

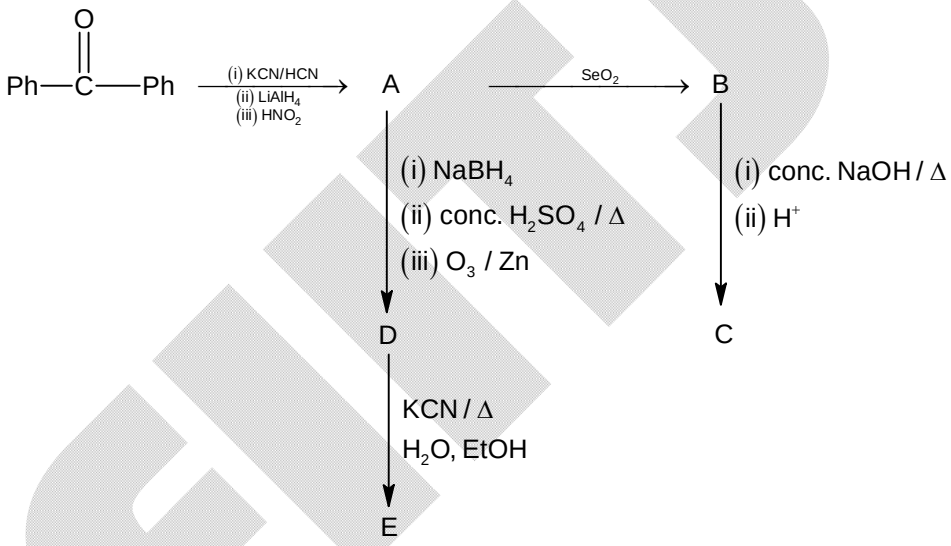
Experiment	[A] (M)	[B] (M)	Initial rate $\left(Ms^{-1}\right)$
I.	0.1	0.1	4×10^{-4}
II.	0.2	0.2	1.6×10^{-3}
III.	0.5	0.1	1.0×10^{-2}
IV.	0.5	0.5	1.0×10^{-2}

The rate of reaction, when concentration of A and B are 0.2 M and 0.35 M respectively, is $x\times10^{-4}$ Ms^{-1}. The value of 'x' is.

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 31 and 32



From the above sequence of reactions, answer the following questions (consider A, B, C, D and E are major organic products)

31. The degree of unsaturation in compound **C** is

32. The number of sp^2 hybridised carbon atoms in compound **E** is

Paragraph for Question Nos. 33 and 34

Buffer solution is a solution of weak acid its conjugate base or a weak base and its conjugate acid. It resists the change in pH when small amount of strong acid or base is added to it.
Based on above paragraph answer the following questions:

33. In an aq. solution of 100 ml, 0.1 M H_2CO_3, 150 ml. 0.1 M NaOH is added, the pH of the resulting solution is found to be 'x'. Now 50 ml, 0.1 M H_2SO_4 is added in resulting mixture. If the pH of final solution is 'y' then the value of (x – y) is
pK_{a_1} & pK_{a_2} of H_2CO_3 is 6 and 10 respectively.

34. What will be the pH, when 24.4 g of benzoic acid is added at room temperature in 500 mL of 0.1 M NaOH solution (considering no change in volume on addition of benzoic acid in solution).
[Given: pKa of benzoic acid is 4.2 at 25°C. atomic mass C = 12, O = 16, H = 1.
log2 = 0.3, log3 = 0.5, log5 = 0.7]

Mathematics

PART – III

SECTION – A
(One Options Correct Type)

*This section contains **FOUR (04)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is the correct answer.*

35. Let $\theta \in \left(\frac{\pi}{3}, \frac{\pi}{2}\right)$ and a function f(x) is defined as

$$f(x) = \begin{cases} \lim\limits_{n\to\infty} \dfrac{(ax^3 - bx^2)(\sin\theta - \sin^3\theta) - (c - 11x)\left|\sin\theta - \sin^3\theta\right|^n}{(\sin\theta - \sin^3\theta) - \left|\sin\theta - \sin^3\theta\right|^n}, & x \in Q \\ \lim\limits_{n\to\infty} \dfrac{(ax^3 - bx^2)(\sin\theta + \sin^3\theta) + (c - 11x)\left|\sin\theta + \sin^3\theta\right|^n}{(\sin\theta + \sin^3\theta) + \left|\sin\theta + \sin^3\theta\right|^n}, & x \notin Q \end{cases}$$

If f(x) is continuous at x = 1, x = 2 & & x = 3, then the values of (|a| + |b| + |c|) is:

(A) 6 (B) 13
(C) 12 (D) 11

36. Let $\vec{a} = \hat{k} + \hat{j} + 4\hat{k}$ and $\vec{r}$ be a variable vector such that $\vec{r}\cdot\hat{i}, \vec{r}\cdot\hat{j}$ and $\vec{r}\cdot\hat{k}$ are positive integers. If $\vec{r}\cdot\vec{a} \le 14$, then the number of possible $\vec{r}$ vectors is

(A) 61 (B) 78
(C) 364 (D) 245

37. Line segments AK_1, AK_2, ….., AK_n are drawn from A(1, 1) where K_1, K_2, K_3, ….. K_n are points in 1st quadrant on $\frac{(x-1)^2}{a^2} + \frac{(y-1)^2}{b^2} = 1$ (a > b) such that chord AK_r makes an angle of $\frac{r\pi}{2n}$ with the positive x-axis. The value of $\lim\limits_{n\to\infty}\left(\frac{1}{n}\left(\sum\limits_{r=1}^{n}(AK_r)^2\right)\right)$ is

(A) $\frac{\pi}{2}ab$ (B) $\frac{2}{\pi}ab$
(C) ab (D) $\frac{1}{ab}$

38. The number of ways of arranging n boys (B_1, B_2, ... B_n) and n girls (G_1, G_2 ..... G_n) around a circle such that boys and girls should sit alternatively and B_1 and G_1 are not together is

(A) (n – 1)! (n – 2)! (B) (n – 1)! (n – 2)! $(n^2 + n + 2)$
(C) (n – 1)! (n – 2)! $(n^2 + 2)$ (D) (n – 1)! (n – 2)! $(n^2 - 3n + 2)$

SECTION – A
(One or More than one correct type)

*This section contains **THREE (03)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

39. If $f(x) = x^2 - \pi x$ and $g(x) = x^2 - 2\pi x$ and $p = \sin^{-1}\frac{2}{3}$, $q = \cos^{-1}\left(\frac{-1}{2024}\right)$, $r = \tan^{-1}\left(\frac{2024}{2023}\right)$, then

(A) f(p) > f(r) > f(q) (B) g(p) > g(r) > g(q)
(C) g(p) > g(q) > g(r) (D) f(p) < f(r) < f(q)

40. At the point P(a, a^n) on the graph of $y = x^n \ \forall\ n \in I^+$, a > 0, a normal is drawn to intersect the y-axis at point (0, b) and if $\lim_{a\to 0} b = l$, then

(A) l = 0 if n < 2 (B) $l = \frac{1}{2}$ if n > 2

(C) l = 2 if n > 2 (D) $l = \frac{1}{2}$ if n = 2

41. Family of curves which makes an angle of $\frac{\pi}{4}$ with the family of hyperbola xy = a, is

(a > 0, and a is a parameter)

(A) $x^2 - 2xy - y^2 = A$ (B) $y^2 - 2xy - x^2 = A$

(C) $x^2 + 3xy - y^2 = A$ (D) $y^2 + 4xy - x^2 = A$

SECTION – B

(Numerical Answer Type)

*This section contains **SIX (06)** Numerical based questions. The answer to each question is a **NON-NEGATIVE INTEGER VALUE**.*

42. If $\cos\theta_1 + 2\cos\theta_2 + 4\cos\theta_3 = \sin\theta_1 + 2\sin\theta_2 + 4\sin\theta_3 = 0$, then $|\cos 3\theta_1 + 8\cos 3\theta_2 + 64\cos 3\theta_3|$ is equal to _____ (where $\theta_1 + \theta_2 + \theta_3 = \pi$)

43. If A is the matrix $\begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, then $A - \frac{1}{3}A^2 + \frac{1}{9}A^3 + ... + \left(-\frac{1}{3}\right)^n A^{n+1} + ... \infty = \begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}$. Then the value of (|a| + |b| + |c| + |d| + |e| + |f| + |g| + |h| + |i|) _____

44. Let $f(x) = \tan^3 x + a\tan^2 x + b\tan x + c$, where a, b, c are chosen respectively by throwing a dice three times. The probability that f(x) is an strictly increasing function $\forall\ x \in R - (2n + 1)(\pi/2)$ where $n \in I$, is equal to (a/b), where a, b are coprime natural numbers, then b + a is equal to _____

45. Area enclosed by y = g(x), x = – 3, x = 5 and x–axis where g(x) is inverse function of $f(x) = x^3 + 3x + 1$ is A, then [A] is _____ (where [.] denotes the greatest integer function)

46. Let 'L' be a line passing through A (1, 1, 1) and intersects lines x + y = 1, z = 0 and x – y = 2, y + z = 3. If 'θ' be the acute angle between line L and line $\frac{x}{1} = \frac{y}{2} = \frac{z-1}{3}$ and the value of $\cos\theta$ is $\frac{3\sqrt{n}}{\sqrt{14}\cdot\sqrt{8}}$, then [n] is _____ (where [.] denotes the greatest integer function)

47. Let $\frac{1}{a_1 - 2i}, \frac{1}{a_2 - 2i}, \frac{1}{a_3 - 2i}, \frac{1}{a_4 - 2i}, \frac{1}{a_5 - 2i}, \frac{1}{a_6 - 2i}, \frac{1}{a_7 - 2i}, \frac{1}{a_8 - 2i}$ are vertices of regular octagon. If the area of octagon is A, then the value of $8\sqrt{2}A$ is _____
(where $a_i \in R$ for i = 1, 2, 3, 4, 5, 6, 7, 8)

SECTION – C
(Numerical Answer Type)

*This section contains **TWO (02) paragraphs.** Based on each paragraph, there are **TWO (02)** questions of numerical answer type. The answer to each question is a **NUMERICAL VALUE (XXXXX.XX).** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

Paragraph for Question Nos. 48 and 49

Let P(x) be a polynomial with real coefficient such that $(x^2 + x + 1)\,P(x - 1) = (x^2 - x + 1)\,P(x)\ \forall\ x \in R$ and $P(1) = 3$

48. If $\int_0^1 \tan^{-1}\left(\frac{2x}{1+P(x^2)}\right)dx + \int_0^1 \tan^{-1}(x+1)\,dx = \frac{k}{16}(\pi - \ln 4)$ then value of k is

49. Let $I_n = \int_0^1 \{P(x^2) + P(x^2 - 1)\}^n\,dx$. If $(k + 1)\,I_n = 2k\,(1 + I_{n-1})$, then the value of n is $(k, n \in N)$

Paragraph for Question Nos. 50 and 51

A line passing through A(–1, –1) intersects the parabola $y^2 = 4x$ at P and Q. Focus of parabola is S(1, 0) (Where [.] denotes the greatest integer function)

50. If Q is $\left(\frac{9}{4}, 3\right)$ and equation of bisector of angle PSQ is $y = mx + c$ then $[m^2 + c^2]$ is equal to

51. Let PP′ be the latus rectum, QQ′ be a focal chord (P′ and Q′ lie on the parabola). The point of intersection of PQ′ and P′Q is $B(\alpha, \beta)$ then $[\alpha^2 + \beta^2]$ is equal to $(\beta > 2)$